Math 1200 Lecture 15

Oct 27 2026

Topics: 16.3: Conservative vector fields, Fundamental Theorem of Line Integrals

Name:_____

Collaborators:_____

Group work 1 Determine whether each of the following vector fields $\vec{F}$ is conservative, and if so find a potential.

a) $\vec{F} = \langle \sqrt{1+x^2+y^2}, \sqrt{1+x^2+y^2} \rangle$

b) $\vec{F} = \langle 3x^2y + 2y^2, \cos(y) + x^3 + 4xy \rangle$

Group work 2 Calculate the vector line integral $\int_C \vec{F} \cdot d\vec{r}$ and verify using the FTLI.

a) $\vec{F} = \nabla f$ for $f = x^2y$ and C the straight line from $(0,0)$ to $(1,2)$

b) $\vec{F} = \widehat{j}$ and C the unit circle parametrized counterclockwise.

Group work 3 (Discussion) Let $\vec{F}$ be a conservative vector field on an open simply connected region D and let C_1 and C_2 be two paths from (x_0, y_0) to (x_1, y_1). What can we say about $\int_{C_1} \vec{F} \cdot d\vec{r}$ and $\int_{C_2} \vec{F} \cdot d\vec{r}$?

Group work 4 Let $\vec{F} = \langle y+z, x+z, x+y \rangle$ and let C be the path drawn below, oriented from $(1, \frac{1}{2}, 0)$ to $(2, 3, 2)$. Calculate $\int_C \vec{F} \cdot d\vec{r}$.

